

Adi Menon

adimenon.me
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I want to become a highly adaptable mechanical engineer to preserve and defend our natural environments

STANDARDIZED TESTS

SAT
R/W 740, Math 720, 2026

AP
AP Scholar, College Board, 2026
- Calculus BC 3
- Computer Science 3
- English Language 4
- United States History 4

VAPA AWARDS

Halloween Horrors Film Competition, Santa Cruz, CA
- 2nd place, screenwriter/audio technician, 2025
- 4th place, director, 2024

SKILLS

Coding
Proficient in Java, Python, React, TypeScript

Electronics
Advanced in circuit design, model making, soldering

Robotics
Advanced in CAD, circuit design, Manufacturing and testing, PCB design, technical report writing, software development

EDUCATION

Pacific Collegiate School, Santa Cruz, CA — Class of 2027
GPA 4.34 weighted, 3.83 unweighted
12th: AP English Literature, AP Physics C (E&M), AP Microeconomics, AP Statistics, Topics in Design & Engineering (in progress)

11th: AP Computer Science A (A/A), AP English Language (A/A), AP Physics C Mechanics (A/A), AP United States History (A/A), Multivariable Calculus Honors (A/B)

10th: AP Calculus BC (A/A), AP World History (A/A), Chemistry (A/A), English 10 (A/A), AP Spanish (A/B), Video Production 2 (A/A)

9th: ; Art1 (A/A), Biology (A/A), English 9 (A/A), Precalculus Honors (A/A), Spanish 3 (A/A), World History (A/A), Video Production (A/A)

University of North Dakota, und.edu/academics/online — Fall 2026
Math 207: Introduction to Linear Algebra (2 credits, in progress)

Stanford University, Stanford, CA — Summer 2026
CS106EA: Exploring Artificial Intelligence (3 credits, B+)

UC Scout, ucscout.org/courses — Fall 2023
JAVA140: Introduction to Java (5 credits, A)

SUMMER ACTIVITIES

Karr Company , Remote, karrcompany.co.uk — Summer 2025
Software Intern, 5 hours per week, 6 weeks
- Coded in Java and React, built a functional prototype, contributed to both front-end and back-end development, shaped the product's core features and user experience, worked closely with CTO
- AI-powered platform to automate and simplify vehicle administration, fines, and expenses

Inspirit AI, trial.inspiritai.com — Summer 2024
- Course: Artificial Intelligence Intensive for High School students
- Built and implemented real-world image recognition models

AWARDS & HONORS

2026

International Science and Engineering Fair (ISEF), 2026
- $500 1st Award, Patent and Trademark Office Society. Qualified participant, Robotics category, individual, tinyurl.com/AdiM-ISEF

California Science Engineering Fair (CSEF), 2026
- Qualified participant, Robotics category, individual

Santa Cruz County Science Fair, 2026
-Best In Show/ISEF Finalist, senior division
Engineering category. Outstanding Electrotechnology Experiment, IEEE.
Outstanding Project, US Air Force

The Conrad Challenge
-Distinguished Conrad Innovator Award, Founder/Captain, team of 2, *Fireguard Technology*, tinyurl.com/nhzj7nmh

2025

International Science and Engineering Fair (ISEF), 2025
- Qualified as Observer

California Science Engineering Fair (CSEF), 2025
- Qualified participant, Computer Science category, individual

Santa Cruz County Science Fair, 2025
-1st place, Computer Science category, individual,
-Outstanding Electrotechnology, IEEE:

INDEPENDENT ENGINEERING ACTIVITIES

Learn, Research, Build, 5 hours per week, year-round Since 2018
- Self-study: "What If", MAKE magazine, and information about sensors and autonomous vehicles
- Research online: motion sensors, software, AI
- Create and develop original concepts
- Relentlessly prototype and test mechatronic assemblies
- Scan surrounding environment for unsolved engineering challenges
- Obtain mentorship and feedback from experts, science fair judges, and competition feedback, etc.

Design and Development of an Autonomous Wildfire Suppression Firefighting Robot
- Built a functional prototype for a self-directed fire detection and suppression robotic platform
- Programmed an embedded sensor fusion framework leveraging YOLOv8 object recognition
- Engineered heat-resistant structural shielding to maintain system operation within active flame zones
- Integrated machine learning routines for independent pathfinding and decision-making during suppression tasks
- Calculated power budgets to balance suppression hardware demands alongside emergency communication payloads
- Evaluated lightweight, high-potency fire retardants for optimal payload efficiency
- Created scalable mechatronic and software architecture adaptable for structural safety audits and wildfire risk analysis
- Analyzed operational resource constraints experienced by emergency responders during severe wildfire outbreaks
- Consulted with San Jose Battalion Chief Robert Culbertson to assess real-world firefighting requirements
- Project Writeup

Using Deep Learning AI to Enable the Safety of Self-Driving Cars
- Developed a software-based project to compare two Convolutional Neural Network (CNN) models
- Used CNNs to classify road signs for safer autonomous driving
- Analyzed dataset included 3,530 training and 638 testing images across 14 categories
- Compared VGG-16's architecture with ResNet-50's residual network.
- ResNet-50 achieved 77% accuracy, while VGG-16 achieved only 23%
- Identified VGG-16's likely overfit because its 138 million parameters exceeded dataset needs
- Verified that both models maintained similar testing performance, indicating good generalization.

-$100 award Keet Family Award;
-Exceptional Genius Olympiad Award
https://tinyurl.com/AdiM-2025

2024

Thermo Fisher Scientific Junior Innovator Challenge, 2024
- Qualified Participant

California Science Engineering Fair (CSEF), 2024
- Qualified participant, Physics category, individual

Santa Cruz County Science Fair, 2024
-Honorable Mention for Best in Show, 1st place, Physics category;
-US Air Force- Most Outstanding Exhibit in Science, Technology Engineering and Mathematics,
-Yale Science and Engineering Association
-Naval Science Award, Office of Naval Research
-Best in Innovation and Design, Tech Interactive;
-Outstanding Electrotechnology Experiment, IEEE
-Science Achievement Award, California Association of Professional Scientists Award: tinyurl.com/AdiM2024

2023

Thermo Fisher Scientific Junior Innovator Challenge, 2023
- Qualified Participant

California Science Engineering Fair (CSEF), 2023
-3rd place, Physics category, individual

Santa Cruz County Science Fair, 2023
-1st place, Physics Category
-Outstanding Electrotechnology Experiment, IEEE
-Science Achievement Award, California Association of Professional Scientists Award: tinyurl.com/AdiM-2023

2022

Thermo Fisher Scientific Junior Innovator

Towards Sensor Fusion

- Investigated whether one accurate sensor outperforms combined sensors in autonomous vehicles.
- Developed a multimodal sensor fusion project to precisely locate objects in the field of view
- Created a precise location map of surroundings using LiDAR Ultrasonic sensors and Infrared sensors
- Combined digital readouts from the three sensors to create a confirmation of an obstacle on the road
- Calibrated XY location of the obstacle and triangulated location using fused sensor data

Ultrasonic Eyes

- Researched self-driving cars for existing deficiencies and developed a test to understand HC-SR04's [ultrasonic sensor] accuracy
- Investigated how temperature, materials and environmental conditions affect HC-SR04 ultrasonic sensor accuracy
- Tested cotton, sponge, wood, sandpaper, heat, humidity, wind, and sound interference with cotton clearly preventing detection and acoustic tests becoming inaccurate beyond 70 centimeters
- Demonstrated that most conditions minimally affect readings, suggesting stronger external factors were required

Flying High with Drones

- Built a 16-gram basic quadcopter with motors and popsicle sticks to test payload capacity at different voltages

Challenge,2022
- Qualified Participant

California Science Engineering Fair (CSEF), 2022
- Qualified participant, Physics category, individual

Santa Cruz County Science Fair, 2022
-1st place, Physics category
- Outstanding Electrotechnology Experiment, IEEE
-US Metric Association Award
tinyurl.com/AdiM2022

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2026 MATE ROV World Championship, St. John's, Newfoundland & Labrador, Canada
-1st place, team Hephaestus, Ranger category (47 teams, 15 countries)

2026 MATE Northern California Regional ROV Competition, Watsonville, CA
- 1st place, team Hephaestus, Ranger category

2025 MATE ROV World Championship, Alpena, Michigan, USA
-3rd place team Hephaestus, Ranger category

2025 MATE Northern California Regional ROV Competition, Watsonville, CA
- 1st place team Hephaestus, Ranger category

2024 MATE Northern California Regional ROV Competition, Watsonville, CA
- 2nd place ,team Angry Otters , Navigator category

- Added 1–7-gram payloads, measuring five-second flight heights across four trials
- Discovered that adding 9V increased flight height but produced greater variability than six volts
- Concluded that drones could only carry less than half their weight

TEAM ROBOTICS ACTIVITIES

Hephaestus Robotics Team/ Angry Otters, X Academy.org

Santa Cruz, CA, 5-20+ hours per week — Since 2023

VP Engineering, since 2025
- Lead a group of 3 coders, built an AI-based image detection system
- Lead the ROV hardware integration team of 4, support the CTO in making ROV
- Build, test, write technical reports; develop software with team members

Electrical Engineer/Circuit Designer, 2024-2025
- Custom-designed the PCB's for ROVs' internal electronics
- Collaborated with team in technical report writing and circuit design
- Developed an AI recognition software that identifies invasive European green crabs in real time

Circuit Designer, 2023-2024
- Constructed the inner electronics of the ROV, developed custom PCB boards to maximize ROV productivity

HOBBIES

Cooking
- Amateur cook and recipe creator, since 2020

Reading
- Avid reader of fantasy fiction and graphic novels since 2014

Guitar
- Self-educated guitarist, since 2018
- Learn new techniques (finger picking, hammer-offs, etc.) from YouTube
- Practice tablature of rock songs
- Self-entertainment
- Link/ youtube.com/shorts/efPxoNFndTI?si=kuVqUV7NYliKU-zu

Wood Work
- Amateur carpenter/artist, since 2020
- Whittle, sand, carve; figurines
- DIY using Dremel 4000 Rotary Tool

Tennis/Biking…
- Recreational player/enthusiast, since 2019

COMMUNITY/SCHOOL CLUB ACTIVITIES

Aegis Living, aegisliving.com, Aptos, CA Since 2026
30 hours
- Volunteer, support elderly in an assisted living facility with tech needs and recreation

Teen Kitchen Project, teenkitchenproject.org, Santa Cruz, CA
April - December 2023
60 hours
- Volunteer, prepared and cooked nourishing meals for families facing cancer and other life-threatening illnesses in Santa Cruz County

YouthServe, scvolunteercenter.org, Santa Cruz, CA June- August 2024
60 hours
- Volunteer, environmental clean-ups, graffiti removal, native plant restoration, beach clean-up, food sorting and packaging, Second Harvest Food Bank, Watsonville, CA

Pacific Collegiate School pacificcollegiate.com, Santa Cruz, CA since 2025
80 hours
- President, Electronic Club, 10-13 members, various electronics projects/repairs, study electrical engineering, since 2025
- Member, Philosophy Club, ~10 members, weekly discussion on humanity and other topics, 2025
- Member, Ultimate Frisbee Club, 30 members, biweekly practice, 2023-2024
- Volunteer, various community and school events: State Park habitat restoration, Iron Man triathlon, school orientation leader for new student initiation, beach cleanup, tutoring math, Stuff the Bus, video recording for 4 school events, food sorting and packaging at Grey Bears